Weighted and Generalized Least Squares

Applied Linear Models I • Lecture 14 of 16

Applied Linear Models I

Today

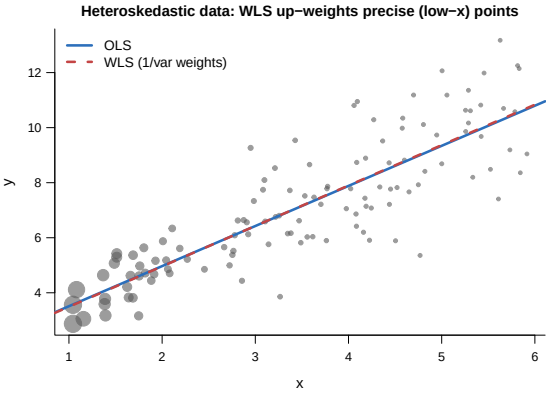
When errors are **non-spherical** ($\mathrm{Var}(\varepsilon) \neq \sigma^2 \mathrm{I}$):

- **WLS** for heteroskedasticity (diagonal covariance);
- **GLS** (Aitken) for any covariance;
- **FGLS** when the variance must be estimated;
- WLS vs OLS + robust SE – efficiency vs robustness.

Weighted least squares

Independent but heteroskedastic errors, $\mathrm{Var}(\varepsilon_i) = \sigma_i^2$:

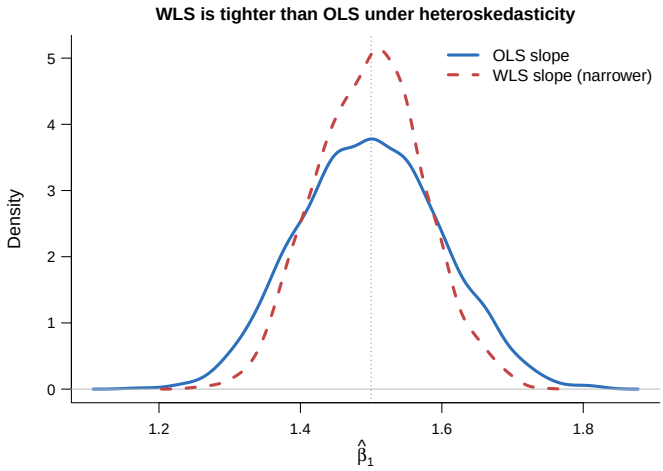
$$\hat{\boldsymbol{\beta}}_{\mathrm{WLS}} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}, \qquad w_i = \frac{1}{\sigma_i^2}.$$



- Precise points up-weighted.
- "Transform then OLS": rescale by $1/\sigma_i$.
- Matches `lm(weights=)` exactly.

WLS is more efficient

With correct inverse-variance weights, WLS is **BLUE**.



Generalized least squares (Aitken)

General covariance $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{\Omega}$ (unequal variances *and* correlation):

$$\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = (\mathbf{X}^\top \boldsymbol{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Omega}^{-1} \mathbf{y}.$$

Aitken theorem

GLS is **BLUE** under non-spherical errors – the generalization of Gauss–Markov from $\sigma^2 \mathbf{I}$ to $\sigma^2 \boldsymbol{\Omega}$.

Worked example: AR(1) errors ($\rho = 0.7$). OLS unbiased; GLS uses the correlation for efficiency.

Feasible GLS

$\boldsymbol{\Omega}$ is usually unknown $\Rightarrow$ estimate it (FGLS):

1. OLS $\rightarrow$ residuals e_i.
2. Model the variance: regress $\log e_i^2$ on predictors $\rightarrow \hat{\sigma}_i$.
3. WLS with $\hat{w}_i = 1/\hat{\sigma}_i^2$ (iterate if desired).

Worked result: FGLS $\approx$ the infeasible true-weight WLS. Only as good as the variance model.

WLS vs OLS + robust SE

	OLS + HC3	WLS / FGLS
Coefficients	OLS (unweighted)	re-weighted
Goal	valid **inference**	**efficiency**
Needs variance model?	no	yes
Robust to bad variance model?	yes	no

WLS SEs $<$ classical and HC3 here – the efficiency dividend, *if* the weights are right. Not rivals: **different priorities**.

Summary

- WLS $= (X^\top W X)^{-1} X^\top W y$, $w_i = 1/\sigma_i^2$; transform then OLS.
- WLS is BLUE and much more efficient under heteroskedasticity.
- GLS (Aitken) handles any covariance; FGLS estimates it.
- WLS for efficiency (given a variance model); robust SE for safe inference.

Next

Lecture 15: robust, nonparametric, and shrinkage estimation.